

1 Introduction

Team Street Precision Las Vegas has successfully transitioned from basic hardware verification to a functional autonomous driving pipeline. Our vehicle now demonstrates closed-loop lane following and speed regulation around 25 cm/s. This phase focuses on integrating the Jetson Nano as the main compute platform and refining perception and control to align with Bosch Future Mobility Challenge requirements for autonomous navigation in a miniaturized smart city.

[Link to multimedia file.](#)

2 Planned activities

A1: Lateral Control (Lane Following): Implement an OpenCV-based lane detection pipeline (Canny + Hough / similar) and feed the resulting lane-center error into the steering controller to achieve autonomous lane following on straight segments and curves.

A2: Longitudinal Control (Speed): Implement and tune a feedback speed controller using encoder data, including “city” and “highway” speed profiles, so the vehicle can hold ~25 cm/s in technical-run conditions and automatically slow down on curves.

A3: Autonomous Orchestration: Merge perception and control into a single closed-loop pipeline on the Jetson Nano, define a fixed control rate, and measure end-to-end latency from camera frame acquisition to actuator command to stay within 10–20 Hz operation.

A4: AI Hardware & Inference: Integrate the Jetson Nano as the main compute unit and bring up the AI stack, including initial YOLOv5 experiments for detecting signs, pedestrians, and traffic lights in preparation for later decision-making.

A5: Physical Platform Maintenance: Improve mechanical and electrical reliability of the vehicle (friction reduction, cable/SD-card robustness) to ensure stable behavior during longer autonomous test sessions.

3 Status of planned activities

Activity 1: Lateral Control (Lane Following)

- Status: Completed.
- Details: An OpenCV pipeline using Canny edge detection and Hough-style line extraction is in place to estimate lane center. This lateral error feeds the steering controller, allowing the car to track straight segments and moderate curves on the test track.

Activity 2: Longitudinal Control (Speed)

- Status: Completed at baseline; multi-profile logic under debugging.
- Details: A feedback speed controller using encoder data is implemented and tuned around a 25 cm/s target consistent with BFMC technical runs. A city/highway speed-profile logic has been coded to reduce speed in curves and allow higher speeds on straighter sections; this profile switching is currently being debugged and validated.

Activity 3: Autonomous Orchestration

- Status: Completed.

- Details: Perception and control have been merged into a single pipeline running at a fixed control rate on the Jetson Nano. End-to-end latency from camera frame acquisition to actuator command is being profiled, with a working target compatible with 10–20 Hz closed-loop operation.

Activity 4: AI Hardware & Inference

- Status: Ongoing.
- Details: The Jetson Nano is mounted and network connectivity is established. Initial YOLOv5 experiments have started using Roboflow-based datasets, and integration of object detections (signs, pedestrians, lights) into the vehicle's decision logic is planned for the next iteration.

Activity 5: Physical Platform Maintenance

- Status: Completed.
- Details: Mechanical friction has been reduced and storage robustness improved (e.g., securing the SD card), increasing run reliability during longer test sessions.

4 General status of the project

The project is on track for the February 2nd demonstration, where BFMC expects a clear link between sensor input and vehicle actuation (e.g., camera to steering and speed control). The physical test track is fully set up, and the simulation environment is configured with ROS parameters matching the real vehicle. The car can currently drive straight and navigate basic curves under closed-loop control; the object-detection pipeline is in early integration, and high-level state-machine logic for intersections and other tasks is the next focus.

5 Upcoming activities

- Sim-to-Real Transfer: Calibrate and deploy lane-following and speed-control parameters validated in simulation onto the Jetson Nano-based vehicle, verifying closed-loop performance on the real track.
- Logic Implementation: Implement and tune a Finite State Machine (FSM) to orchestrate behaviors for STOP signs, crosswalks, traffic lights, and lane-keeping, consistent with BFMC task and priority structure.
- Hardware Expansion: Finalize integration of a secondary camera (rear or alternative view) to support more reliable parking and complex maneuvers later in the project.
- Compute Allocation: Define how perception and control loads are split between the Jetson Nano and the existing embedded electronics (Raspberry Pi/Nucleo stack), ensuring that all algorithms meet the required update frequencies for BFMC runs.



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